



HR Attrition Analytics

A complete Data Analyst workflow — SQL analysis, KPI generation, and Power BI dashboard visualization — applied to a 1,470-employee dataset to uncover workforce attrition patterns and drive smarter retention strategies.

Dataset Overview

The dataset captures a comprehensive view of each employee across **35 attributes**, enabling multi-dimensional attrition analysis.

Demographics

Age, gender, education field

Work Profile

Department, job role, overtime

Satisfaction

Job, environment, work-life
balance

Compensation

salary hike %,

Key Workforce KPIs

1,470

Total Employees

1,233

Active Employees

237

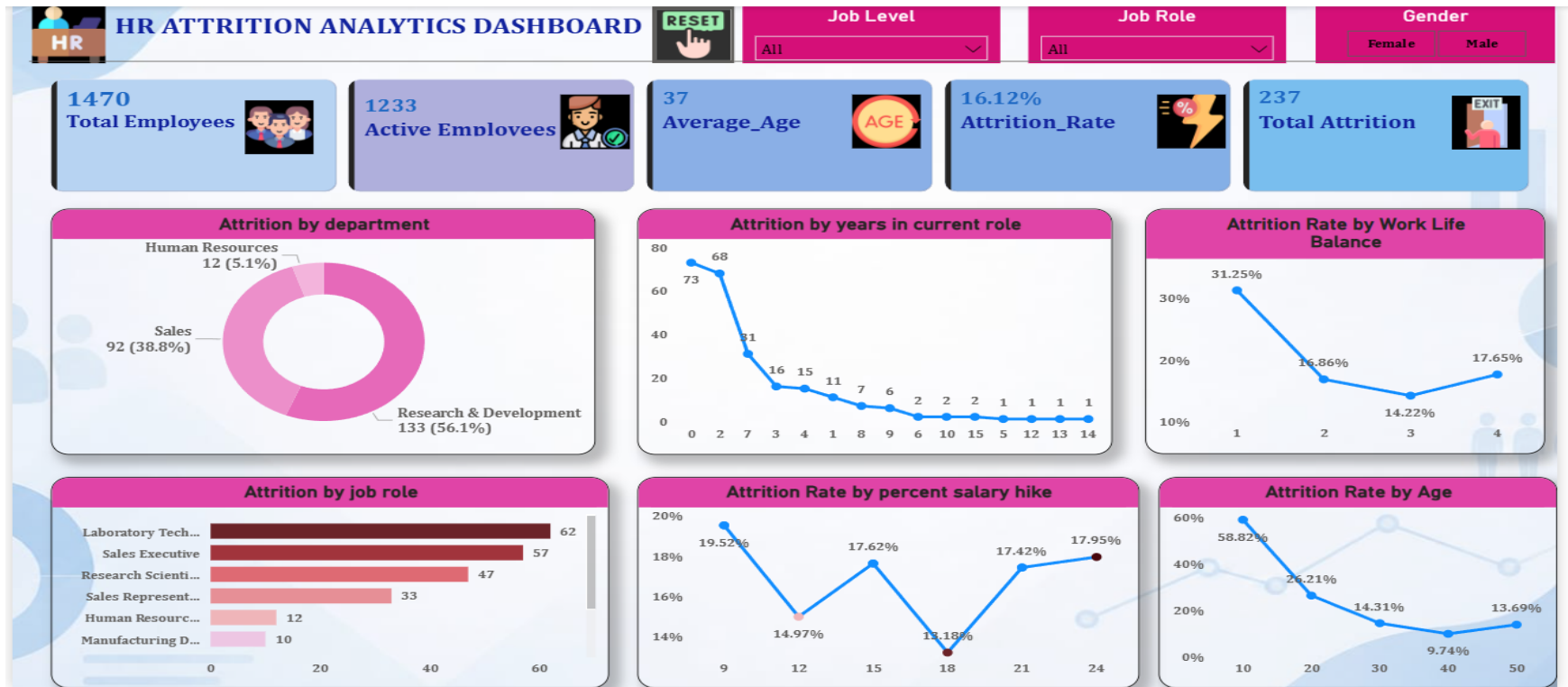
Total Attrition

16.12%

Attrition Rate

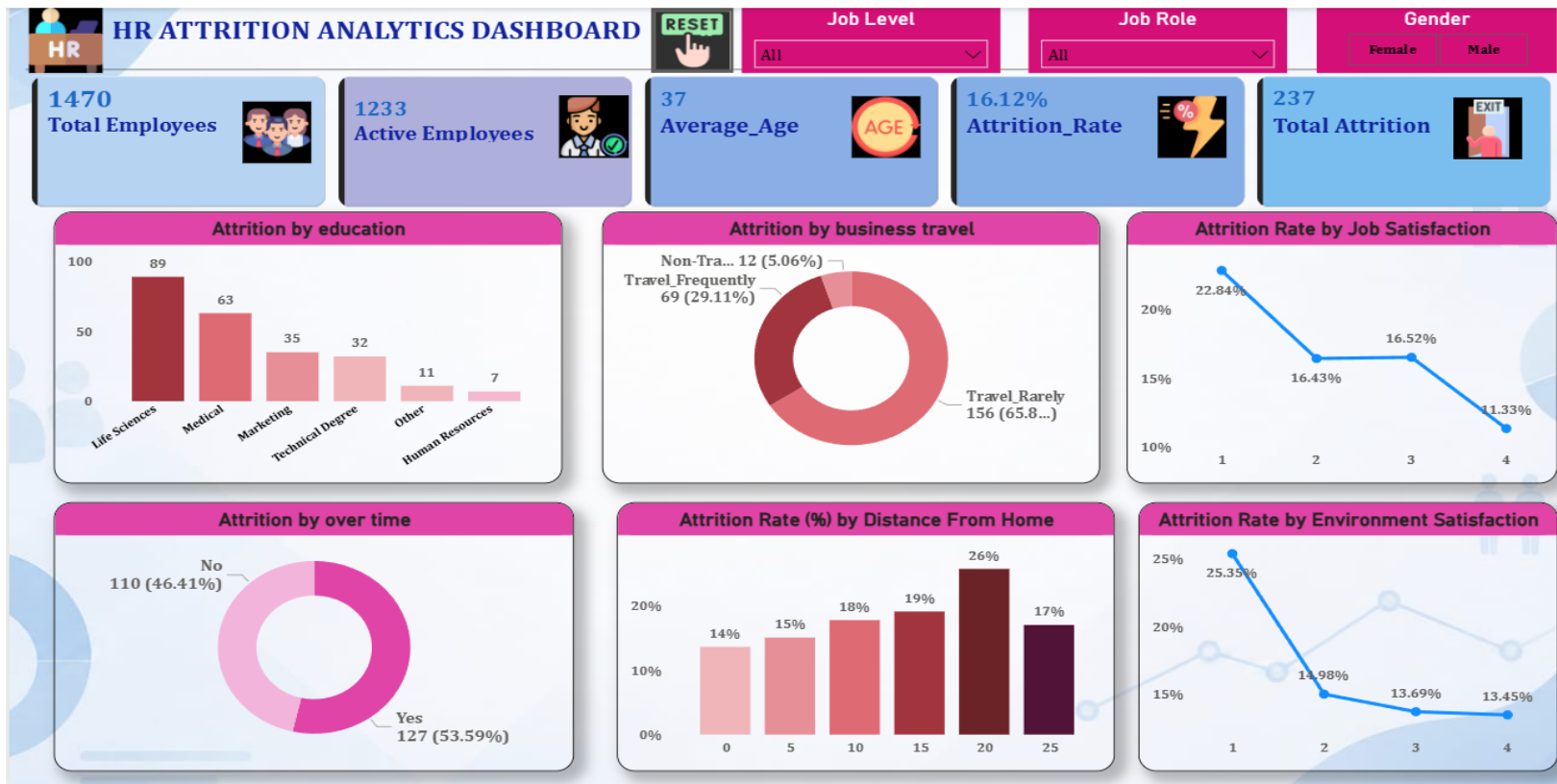
37

Average Age



Dashboard 1

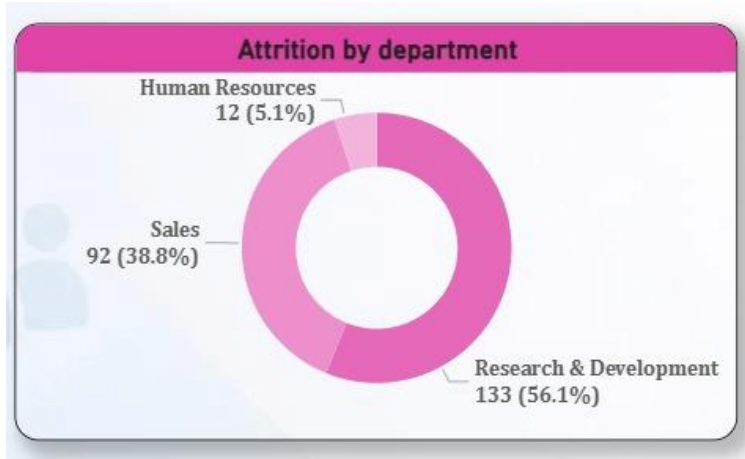
An interactive HR Attrition Analytics dashboard was built in Power BI to visualize attrition trends, department-wise breakdowns, and workforce KPIs – with filters for Job Level, Job Role, and Gender.



Dashboard - 2

The second panel covers attrition by education, business travel, overtime, distance from home, job satisfaction, and environment satisfaction — providing a 360° view of retention risk factors.

Attrition by Department, Role & Age



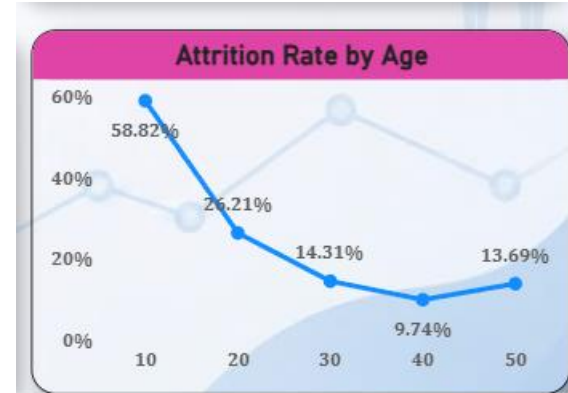
Business Insight

Research & Development department has the highest employee attrition rate.

Business Impact

High attrition in R&D may reduce innovation efficiency and delay project delivery.

Attrition by Age Group



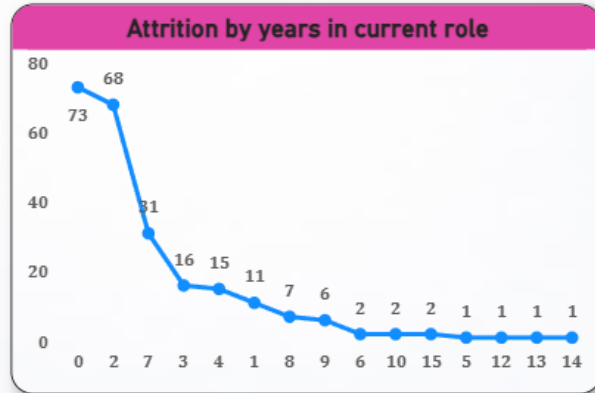
Business Insight

Younger employees show higher attrition compared to older age groups.

Business Impact

High attrition among younger employees may affect long-term workforce stability.

Role Tenure, Work-Life Balance & Salary Hike

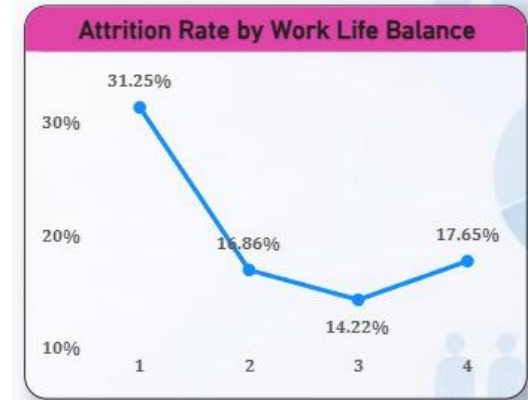


Business Insight

Employees with fewer years in their current role are more likely to leave the company.

Business Impact

Higher early-stage attrition increases hiring and employee training costs.



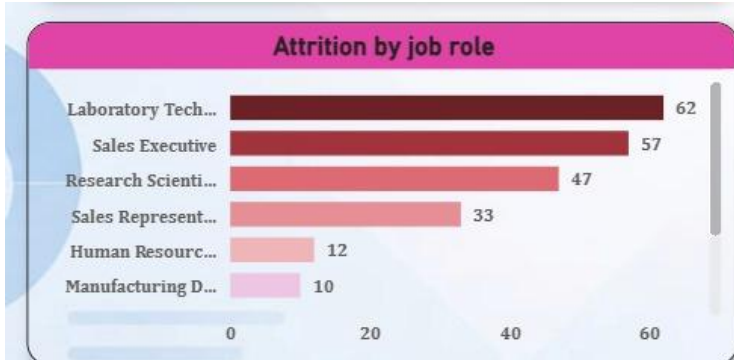
Business Insight

Employees with poor work-life balance show significantly higher attrition rates.

Business Impact

Poor work-life balance can lead to employee burnout and lower productivity.

Job Role, Education & Business Travel

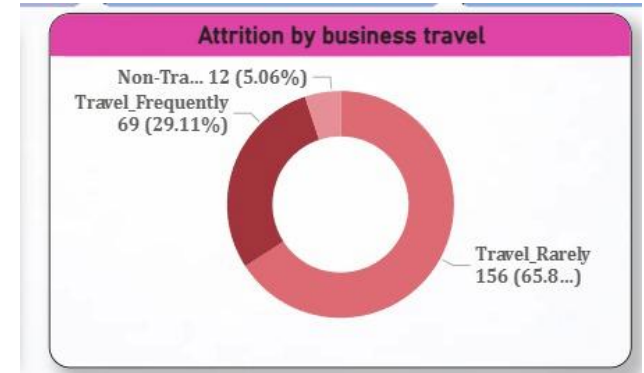


Business Insight

Laboratory Technicians experience the highest attrition among all job roles.

Business Impact

Losing specialized employees can negatively impact operational efficiency.



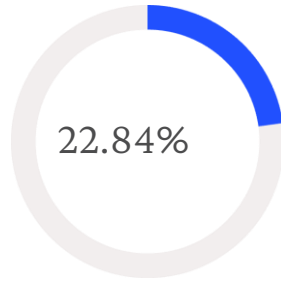
Business Insight

Employees who travel frequently are more likely to leave the company.

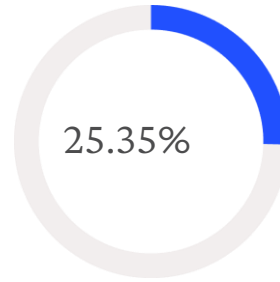
Business Impact

Frequent travel requirements may increase employee stress and burnout.

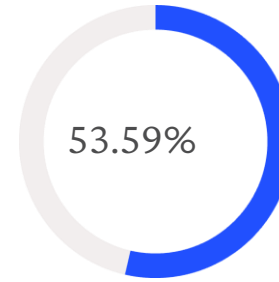
Satisfaction, Overtime & Distance Drivers



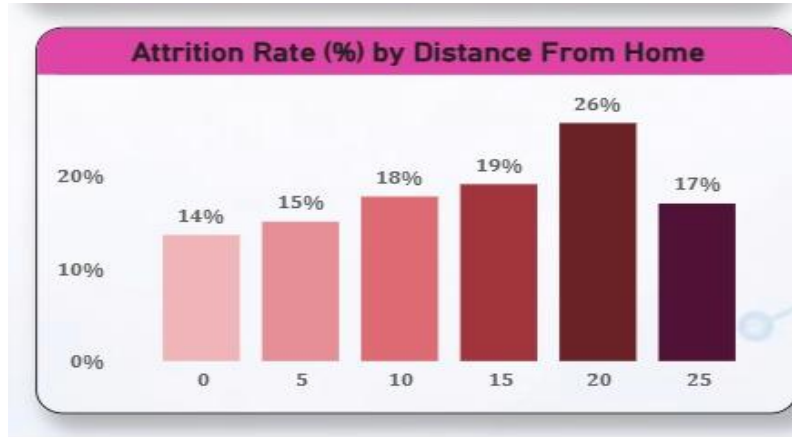
Low Job Satisfaction



Low Env. Satisfaction



Overtime Workers



Business Insight

Higher distance from office leads to higher attrition.

Business Impact

Remote work options can help reduce employee turnover.



Key Insights & Recommendations

Top Insights

- R&D has the highest departmental attrition (56.1%)
- Youngest employees and new-role hires leave most
- Overtime, poor WLB, and low satisfaction are top risk factors
- Distance from home and frequent travel amplify turnover

Recommendations

- Strengthen R&D retention & Lab Technician programs
- Offer competitive salary hikes and career growth paths
- Reduce overtime; implement hybrid/remote work policies
- Boost engagement for younger employees via feedback loops